

Contact

2066362047 (Home)
avanibajaj01@gmail.com

www.linkedin.com/in/avanibajaj
(LinkedIn)

Top Skills

Large Language Models (LLM)
cohere
Semantic Search

Languages

English (Native or Bilingual)
Hindi (Native or Bilingual)
Tamil (Elementary)

Certifications

Machine Learning by Stanford
University
Pretraining LLMs
Masters Program (Master)
Large Language Models with
Semantic Search

Avani Bajaj

Applied Scientist @Amazon | M.S. Data Science @UW| National
Math Olympiad Champ | Green Card EAD
United States

Summary

As a Senior Data Scientist at Tiger Analytics, I collaborate with global clients to analyze large-scale unstructured data and develop data-driven solutions for various domains, such as retail, conversational AI, and autonomous driving. With over 8 years of hands-on experience, I have a strong background in computer science, advanced modeling, and statistical analysis.

I have successfully implemented machine learning techniques, such as predictive modeling, natural language processing, and statistical models, to deliver actionable insights and optimize business outcomes. For example, I increased customer engagement and sales quantity for a retail company by 1.7% using a targeted attribution model. I also improved customer experience for a ChatBot product by 2% using a semi-supervised machine learning model. I am proficient in data modeling, statistical analysis, and data visualization tools, such as Tableau and PowerBI.

I graduated with a Master's degree in Data Science from the University of Washington, where I worked on a capstone project with Microsoft to predict high-potential startups using hierarchical clustering and classification models. I achieved a true positive rate of 52% from 10% on a random sample, and my solution was adopted by the Microsoft team to invest in early-stage startups. I also hold a certification in Machine Learning from Stanford University and a Bachelor's degree in Computer Science from PSG College of Technology.

I am passionate about translating complex business challenges into data-driven solutions, and I have a keen interest in research and innovation. I have published multiple papers in reputed journals and conferences, and I have filed multiple patents in the field of machine learning and data science. I have excellent communication skills, rapid adaptability to emerging technologies, and a collaborative

mindset. I am an outcome-oriented data scientist, motivated to provide valuable insights and drive business growth.

Experience

Amazon

Applied Scientist

August 2025 - Present (11 months)

Sunnyvale, CA

Intuit

GenAI Data Scientist

August 2024 - August 2025 (1 year 1 month)

Mountain View, CA

Stealth Startup

Founding Engineer/ GenAI Data Scientist

April 2024 - April 2025 (1 year 1 month)

San Jose, CA

Tiger Analytics

Senior Data Scientist

April 2021 - July 2024 (3 years 4 months)

United States

Discover - enhanced credit risk assessment with hypothesis testing | AWS, MySQL

- Optimized credit underwriting by testing and updating fraud rules through A/B experiments; overall false positive rate decreased by 2%.
- Designed and executed a comprehensive data pipeline on AWS to capture, process, and analyze customer data. Developed SQL queries to assess creditworthiness and identify high-risk customer segments. Achieved a 15% reduction in loans disbursed to high-risk customers while maintaining a 10% decrease in default rates, demonstrating the effectiveness of data-driven risk management strategies.

Charles Schwab - conversational AI | Python, TensorFlow, DialogFlow, Tableau, MySQL

- Built semi-supervised ML model to personalize chatbot query suggestions; 200K interactions (+2%) using smart prompts within a quarter.

- Designed and implemented A/B experiments to identify optimal model threshold for intent selection in ChatBot product using Dialogflow; overall false positive rate decreased by 2% (350K interactions monthly).

Gap Inc - operations optimization | PySpark, Azure Management, Databricks, PowerBI

- Developed a targeted attribution model analyzing user-click behavior during checkout - leveraged to tailor marketing strategy leading to a 1.7% increase in customer engagement (30M+ users) in US & Canada.
- Engineered and optimized a scalable data pipeline on Azure to process user interaction data, enabling Monte Carlo simulations across 500+ parameter combinations that improved recommendation engine performance by 15% CTR and 4% conversion.

Microsoft

AI/ML Consultant

September 2020 - March 2021 (7 months)

Seattle, Washington, United States

- Constructed a semi-supervised machine learning model-based solution to help predict the Start-up with strong growth potential
- Identified the top 1K startups using hierarchical clustering and classification model on 110K data points fetched from Crunchbase
- Achieved a true positive rate of 52% from 10% on random sample; Adopted by Microsoft Team to invest in early-stage startups

Playment

Intern

July 2020 - September 2020 (3 months)

India

- Built a multi-class classifier for instance segmentation for an autonomous driving use case
- Leveraged CNN, Mask R-CNN, and pre-trained weights on MS COCO datasets for pre-labeling objects and boundaries
- This led to 50% increase in the efficiency of annotators while doubling the throughput for the client.

University of Washington

Research Assistant

January 2020 - September 2020 (9 months)

Seattle

- Built a 3-layer hierarchical classifier model to classify warranty complaints across 100+ categories (currently performed manually)
- Trained a model with 4M+ records using lemmatization, TFIDF, glove method, and BERT for pre-processing, and feature extraction
- Achieved an F1-score of 72% and a 50% decrease in the processing time.

GAIN Credit (formerly Global Analytics)

3 years 8 months

Senior Analyst

January 2018 - August 2019 (1 year 8 months)

Noida Area, India

- Instalment Loan Strategy
 - Developing the architecture of Instalment Loans product, offering EMI loan for 3 to 12 months to eligible customers.
 - Implementing the strategy to offer loan amount up to 3000GBP with similar default rate by reassessing customer behavior
- Optimization for lending portfolio
 - Investigated the behavior of customers who opt for a partial temporary arrangement or Payment Holiday on multiple loans, default on their loans and end in debt.
 - Restricted the potential arrangement customer from entering into debt by having additional checks and offering them lower amount
- Price Progression Test
 - Build financial model to assess the chances of a customer to return and apply for another loan.
 - Proposed campaigns with various lower interest rate for customers to increase the return rate by 5%.

Analytics Scientist II

April 2017 - December 2017 (9 months)

Noida Area, India

- Behavior Model Project
 - Generated customer insights by understanding lender characteristics, assimilating parameters from different databases and building multiple models to predict customer behavior.
 - Optimized risk segregation that led to improved portfolio balance and operating margin.
 - Achieved 3000 additional bookings/month (5% of returning portfolio) with similar default rate, and 9,000 GBP reductions in monthly data cost.
- Cycle Of Debt Analysis

- Identified customers who could pay a sufficient amount for their present loan and thus take recursive/multiple loans of higher amount by paying minimum due of the month and end in cycle of debt.
- Recommended actions to prevent customers from ending in debt that they couldn't payback with current cashflow.
- Underwriting optimization for lending portfolio
 - Enhanced underwriting capabilities for various products by proposing, evaluating and implementing improved rules based on analysis of historical data.
 - Assessed business impact of incorporating the proposed modifications, and codified to enhance underwriting capabilities for various financial products.
 - Achieved ~1000 additional bookings/month (2% of returning portfolio) with similar default rate.
- Incremental Loan Amount strategy
 - Architected and implemented end-to-end functionality to proactively reassess customer credit status and offer variable incremental loans to selected customers.
 - Achieved 10% decrease in offered loan amounts in customer segment with higher default rates, and 5% increase in loans for low default rate customer portfolio

Analytics Scientist I

January 2016 - April 2017 (1 year 4 months)

Noida Area, India

- Marketing Campaigns Analysis
 - Uncovered critical trends from marketing campaign data, identifying campaign activities that impact portfolios.
 - Recommended features to focus on for better performance of upcoming campaigns, improving marketing strategy.
- Analyze customer behavior to manage risk and optimize underwriting
 - Built new predictive rules and variables for identifying default behavior of customers..
 - Achieved ~5% decrease in returning customers default rate.
 - Responsible for maintaining credit risk for returning portfolio of LendingStream product–company's premier short-term loan product in the UK.

Global Analytics

Intern - Analytics

June 2015 - December 2015 (7 months)

- Responsible monitoring the Behavior Model and Weekly product portfolio on a regular basis.
- Created and implemented new rules to increase the number of customers and reduce the portfolio risk and fraud, leading to a better portfolio; led to reduction of fraud rate from 0.3% to 0.2%.
- Undertook various analyses to identify customer behavior and implemented rules to remove bad-performing segments of the product; reduced portfolio default rate by 10%.

Acidaes Solutions Private Limited
 Software Engineering Intern/ Trainee
 April 2014 - September 2014 (6 months)
 Noida

Project: Contributed to the development of a Winforms voting application and website maintenance for HDFC Bank. Demonstrated proficiency in .NET, C#, and SQL while collaborating with a team to deliver a functional and user-friendly solution.

Technical Skills and Responsibilities:

- UI Development: Designed and implemented a Winforms application for voter registration and ballot casting, showcasing proficiency in user interface development.
- Database Management: Utilized SQL to create and manage databases for storing voter and candidate information, ensuring data integrity and efficiency.
- Web Development: Collaborated with the HDFC Bank team to maintain and update their website, demonstrating adaptability to different project requirements.
- Backend Development: Developed backend logic for the voting application using C#, connecting the user interface to the database and managing voting processes.
- Testing and Quality Assurance: Conducted thorough testing to identify and resolve issues, ensuring the application's reliability and performance.

Project Outcomes:

- Successfully delivered a functional Winforms voting application meeting project requirements.
- Improved the efficiency of database operations through optimized SQL queries.
- Contributed to the maintenance of the HDFC Bank website, ensuring a positive user experience.

- Demonstrated strong problem-solving and analytical skills throughout the project.
- Successfully collaborated with team members to achieve project goals within the specified timeframe.

Education

University of the Cumberland

Doctor of Philosophy - PhD · (April 2024 - 2029)

University of Washington

Masters, Data Science · (September 2019 - April 2021)

PSG College of Technology

Bachelor of Engineering (B.E.), Computer Science